

SECTION A

1 (short)

(a) Find the limit as $x \rightarrow 0$ of the expression $f(x) = (x \cos x - \sin x)/x^3$. [5]

(b) Find the limit as $x \rightarrow -4$ of the expression $g(x) = \sin(\pi x)/(x^2 - 16)$. [5]

2 (short)

(a) On an Argand diagram show all the solutions of the equation $z^6 = 64$. [5]

(b) On an Argand diagram sketch representative contours of $\text{Im}(z^2) = \text{constant}$. [5]

3 (short)

(a) Show that the three vectors $\mathbf{a} = [3, -3, 0]^T$, $\mathbf{b} = [2, 2, -1]^T$, and $\mathbf{c} = [1, 1, 4]^T$ are orthogonal, and hence find the volume of the box of which they form the sides. [5]

(b) Order the three vectors to make a right-handed set, i.e. bearing the same relationship to each other as the x , y and z axes in Cartesian coordinates. [5]